END TERM EXAMINATION
(CBCS)(SUBJECTIVE TYPE)
B. Tech CSAI, 2nd Semester
(AUGUST, 2022)

Subject Code: BAI - 106	Subject: Database Management Systems
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
	(a)Discuss the importance of entity & referential integrity rules. How are these rules enforced in databases?	
	(b)Differentiate between Inner Join and Outer Join.	
	© Distinguish strong entity set with weak entity set?	
	(d)Convert the following SQL query into an equivalent relational algebra expression: Select A1,A2....An From r1, r2....rm	
	(e)Consider the relation scheme R = {E, F, G, H, I, J, K, L, M, N} and the set of functional dependencies {{E, F} -> {G}, {F} -> {I, J}, {E, H} -> {K, L}, K -> {M}, L -> {N} on R. What is the key for R?	
	(f) Formulate a Concurrency Control Protocol based on Read and Write Locks.	
	(g) Differentiate between Primary Key, Candidate Key and Super Key *by taking examples.*	
	(h)Explain the importance of Defining Views.	
	UNIT-I	
Q2	*(a) Why would one choose a database system instead of simply storing data in regular files? When would it make sense not to use a database system?* (b) Differentiate between Physical and Logical Data Independence	**(10)**
Q3	Draw a labelled E-R diagram for creating a bank database, each bank can have multiple branches & each branch can have multiple Accounts and Loans. Every customer of the bank must have at least one account but is restricted to at most two loans at a time. State clearly any additional assumption you make.	**(10)**
	UNIT-II	
Q4	Consider the following database and answer the queries in relational algebra expression and SQL: Student (roll no, name, address, father_name, dob, email_id. dno) Department (dno, dname, HOD) Result (roll no, subj 1_ marks, sub2_marks, sub3_marks) a) Retrieve name, address & email_id of students having less than 10 marks in sub2. b) Find the total number of students in each department. c) Find the rollno of students those have scored the same marks in sub3	**(10)**

	as scored by " Pooja Sethi". d) Increment the marks of rollno 10 in sub2 by 3	
Q5	(a) Discuss the ACID properties of a Transaction? (b)What do you mean by Union of two Relations? Explain when two relations are said to be Union Compatible?	**(10)**
	UNIT-III	
Q6	Consider the relational schema for published books: BOOKS (Book-title, Authorname, Book- type, list price, Author-Affiliation, Publisher) Suppose the following dependencies exists Book-title ->Publisher, Book-type Book type ->list price Authorname->Author-Affiliation Apply normalization and try to get highest normal form (which is possible) State the reason behind each decomposition.	**(10)**
Q7	What is a serializable Schedule? How it differs from a Serial Schedule? Explain with an Example.	**(10)**
	UNIT-IV	
Q8	(a)Write short notes on following: (i) What is Concurrency Control? Why it is Important? (ii) Various Measures of Query Cost?	**(10)**
Q9	(b) What are the features of Object oriented Database Management Systems? How they are different from RDBMS.	**(10)**